MIL-I-85080/1A(AS)
20 April 1983
SUPERSEDING
MIL-I-85080/1(AS)
28 November 1977

MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, SHRINKABLE WITHOUT HEAT, POLYVINYL CHLORIDE, FLEXIBLE NON-CROSSLINKED

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Department and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this specification and the isssue in effect of MIL-I-85080.

REQUIREMENTS:

Continuous operating temperature range: Class 1: -20°C (-4°F) to +105°C (221°F); Class 2: -20°C (-4°F) to +125°F); Class 3: -65°C (-85°F) to +90°C (195°F)

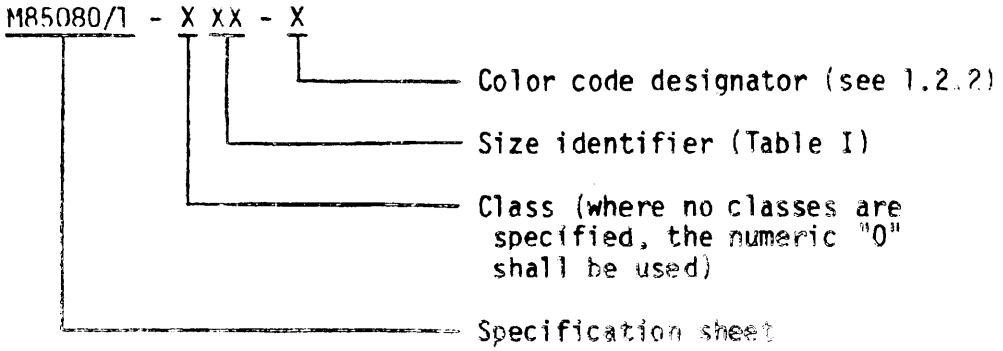
Mode of shrinkage: The heatless shrink tubing shall recover to a predetermined diameter upon exposure to the atmosphere. No external sources of heat are required.

Classes: Class 1-Normal temperature operation
Class 2-Extended temperature operation
Class 3-Low temperature operation

Color: The sleeving shall be furnished in black, white or clear as standard colors. Other colors are available. Colors shall conform to Class II of MIL-STD-104.

Longitudinal change: +2, -10%

Military part number: The military part number shall consist of the basic number of this specification sheet and dash numbers as shown below:



Example: Class 1, black, 1.00 inch as supplied ID sleeving shall be identified as M85080/1-109-0.

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TABLE I. Dimensions (inches). 1/

Military part number 4/	As supplied I.D. min 2/	After unrestricted shrinkage	
		I.D. max	Wall thickness 3/
M85080/1-001-X	.05	.031	.012 ± .003
M85080/1-002-X	.10	.062	.020 ± .003
M85080/1-003-X	.15	.093	.020 ± .003
M85080/1-004-X	.20	.125	.025 ± .003
M85080/1-005-X	.30	.187	.025 ± .003
M85080/1-006-X	.40	.250	.025 ± .003
M85080/1-007-X	.60	.375	.030 ± .005
M85080/1-008-X	.80	.500	.030 ± .005
M85080/1-009-X	1.00	.625	.035 ± .005
M85080/1-010-X	1.20	.750	.035 ± .005
M85080/1-011-X	1.60	1.000	.040 ± .005
M85080/1-012-X	2.00	1.250	.045 ± .005
M85080/1-013-X	2.40	1.500	.050 ± .006
M85080/1-014-X	2.80	1.750	.050 ± .006
M85080/1-015-X	3.20	2.000	.055 ± .007

1/ Diameter limits for object to be enclosed should be in accordance with the recommendations of the manufacturer.

2/ The inner diameter of the sleeve may be mechanically expanded 2.5X the recovered inside diameter when in the as-supplied condition. The manufacturer should be consulted for special applications and recommendations.

3/ Wall shrinkage will be less if shrinkage is restricted.

4/ See Military part number for class designations. Color code identified shall be replaced by the color code from 1.2.2.

Unrestricted shrinkage: 4.6.3.13; 23° ± 1°C(73.4° ± 2°F) for 24 hours.

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TABLE II. Properties after unrestricted shrinkage.

Property	Requirement	Test condition
Restricted Shrinkage	No cracks	4.6.3.2
Voltage Withstand	Pass	4.6.5
Dielectric strength, volts per mil min		4.6.2; ASTM D 876
Wall thickness of:		
.011 to .013	1,150	
.014 to .016	800	
.017 to .024	700	
.025 to .029	630	
.030 to .034	570	
.035 to .039	520	
.040 to .044	480	
.045 to .049	450	
.050 to .054	430	
.055 to .069	400	
Volume resistivity, ohms - cm, min	10^{11}	
Water absorption, %, max	1.0	4.6.2; ASTM D 570, Par 6.1
Fungus resistance	Non-nutrient	4.6.2; ASTM 21
Low temperature flex	No cracking	4.6.6; -10° ± 1°C (+14° ± 2°F)
Heat shock	No cracks, flowing or dripping	4.6.7; Class 1: 180° ± 2°C (356°±2°F Class 2: 200° ± 2°C (392°±4°F); Class 3: 160° ± 1°C (320°±4°F)
Properties after heat test		4.6.8; 130°±2°C (266°±4°F) for 400 hours. Class 2: 150°±2°C 302°±4°F) for 400 hours. Class 3: 115°±1°C (239°±2°F) for 400 hours.
Ultimate elongation, %, min	130	
Ultimate elongation, min. % retention of original value	65	
Corrosion	No corrosion	4.6.9.1, 136°±2°C (276°±4°F) for 168 hours and 4.6.10.3
Properties after fluid resistance test		4.6.10
Tensile strength, psi, min	1,800	
Dielectric strength, volts per mil, min.		
Wall thickness of:		
.011 to .013	805	
.014 to .016	560	
.017 to .024	490	
.025 to .029	440	
.030 to .034	400	
.035 to .039	365	
.040 to .044	335	
.045 to .049	315	
.050 to .054	300	
.055 to .069	280	

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TABLE II. Continued

Property	Requirement	Test condition
Odor	No objectionable odor	4.6.11
Secant modulus, psi, max	15,000	4.6.12.1; 2% strain ASTM D 882
Tensile strength, psi, min	2,000	4.6.13 ASTM D 638, Speed D
Ultimate elongation, %, min	200	4.6.13 ASTM D 638, Speed D
Flammability	Self extinguishing 15 sec or less	4.6.14.2

Shelf Life (as supplied): Test method 4.6.16

Preparing Activity:
Navy - AS
(Project 5970-N596)

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